

8DM60: Multimodal AI in Biomedicine

Lecture 1: Guided self-study (BZ)

Mitko Veta

Department of Biomedical Engineering

Problem 1

A hospital clinical AI team compares two models that draft patient information letters from laboratory reports. On held-out reports, A has lower average next-token loss than B. For this calculation, each token is predicted using the laboratory report and the preceding tokens from the clinician-written reference letter. During generation, each model instead continues from its own previous output.

Yet A more often writes `Your result was normal` when the report says `Result pending` (no result is available). Most reference-letter tokens are shared opening and closing phrases; result statements occupy only a small part.

1. How can A have lower average loss but make more factual errors about results? What does the loss comparison establish about model quality?
2. Would always selecting the most probable next token plausibly solve this problem? Explain.
3. Design a small clinician assessment comparing the generated letters' factual correctness. Specify report selection, what clinicians see and assess, and how you compare A and B.

Problem 2

A biomedical engineering student builds a model to identify which lung contains an opacity (a visible image finding) from a radiology report. Below are two (small) examples of such reports:

- `left opacity right clear`
- `left clear right opacity`

The correct outputs are `left` and `right`, respectively. Compare:

Architecture	Processing steps
A	Token embeddings → add position vectors → Transformer → average the four outputs → classifier
B	Token embeddings → Transformer → add position vectors → average the four outputs → classifier

Each position receives a position vector at the indicated step. Both Transformers allow every token to attend to every token and have no other positional signals or position-dependent operations. Each classifier is a learned linear layer followed by softmax, predicting left/right probabilities. Both models train on labelled examples.

With enough training data and Transformer capacity, can each architecture solve the task? Explain **separately** for A and B, tracing the association between location and finding.

Problem 3

A university biomedical AI researcher trains a two-layer decoder-only Transformer to predict the next token at each position in a radiology report. Training supplies the complete report. The first attention layer accesses every position, only the second has a causal mask.

Three successive input tokens are:

`finding` $\rightarrow$ `is` $\rightarrow$ `absent`

The researcher claims that the second layer's mask prevents the representation at `is` from using `absent` when predicting it.

Is this claim correct? Explain.

Problem 4

A hospital wants a general-purpose language model to generate recommendations following its internal treatment guidelines. These change several times per year and include recommendations specific to rare patient groups poorly represented in training examples.

The searchable current guidelines explicitly cover these groups, but the full collection exceeds the context window. Each request includes the patient characteristics needed to select the applicable recommendation. Answers must identify the supporting document and version.

Propose how the system should obtain and use the relevant guidance. Why would fine-tuning on historical examples alone be insufficient? Describe one way your system could still give an incorrect recommendation.

Problem 5

Machine-learning engineers in three hospital departments train separate LoRA adapters to draft patient letters using the same frozen base model. The central clinical AI team proposes one adapter trained on pooled data. A correct letter preserves input facts and follows departmental instructions.

Department	Letters in evaluation	Separate adapter: correct	Pooled adapter: correct
A	800	90%	94%
B	100	90%	82%
C	100	90%	80%

Evaluation letters are held out from training. The pooled system receives the department name in its prompt. Adapter selection by department is already supported, and both options fit memory. The central team favours pooling for its higher aggregate correctness and single adapter to maintain.

1. Review this recommendation and propose the next action. No exact calculation is required.
2. How can the hospital use three department-specific LoRA adapters without storing three full copies of the base model?